

The ϕ signal: a gradient-free training-stress monitor

Preliminary technical validation note — Simona Vargiu, independent research · July 2026

1. Scope and surviving claim

This note documents the empirical validation to date of ϕ , a training observability signal derived solely from the loss trajectory of a fine-tuning run. ϕ is an exponential moving average (EMA, $\beta = 0.8$) of upward loss jumps, optionally augmented by gradient norm when exposed. It requires no access to gradients, weights, or model internals, and can therefore operate on top of managed fine-tuning APIs (the validation below uses Qwen3-8B via the Tinker API, where only per-step loss is exposed).

The surviving, supported claim is deliberately narrow and should be read as **preliminary** — the evidence base is small (3–12 seeds per test, one model family, one task type): **in the current validation setting, ϕ behaves as a specific, near-instantaneous detector of training instability — not a predictor.** The original hypothesis (an adaptive controller outperforming AdaLoRA) and the predictive-lead-time hypothesis were both tested and are reported in Section 4 as falsified or unresolved.

2. Protocol

All experiments are multi-seed, with decision thresholds fixed *before* data collection and verdict criteria pre-registered in the scripts themselves (including explicit failure conditions). The ϕ alarm threshold used throughout is $\phi = 0.10$, fixed from Test 5 data and reused unchanged in Tests 6–7. Base model: Qwen3-8B; LoRA fine-tuning via Tinker; task: short prompt→target completion pairs; per-step loss logged for every run. Raw traces are provided as JSON files in the accompanying bundle.

3. Supported results (preliminary)

3.1 Regime separation under natural stress (Test 5)

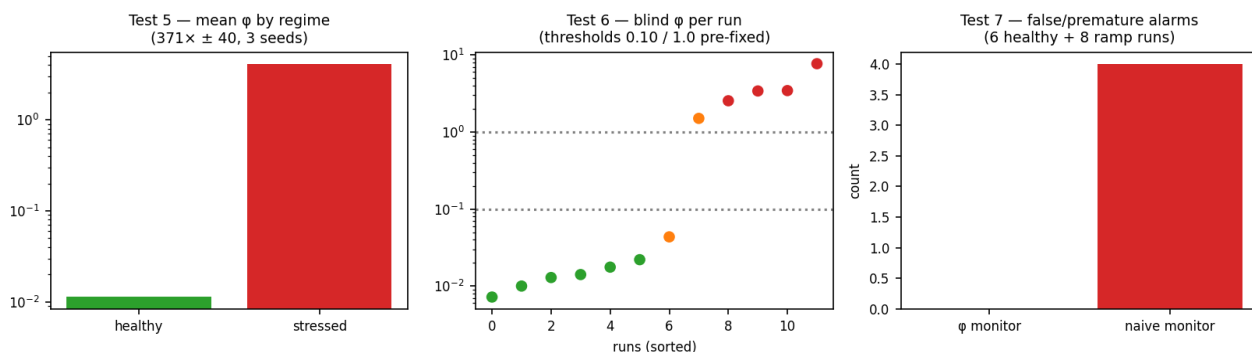
Paired arms on the same task and seeds: healthy LR (1e-4) vs aggressive LR (3e-3), no induced corruption, 3 seeds. Mean ϕ : healthy 0.0113, stressed 4.14 — a separation of **371 $\times$ $\pm$ 40** (per-seed ratios 340 $\times$, 428 $\times$, 345 $\times$), zero NaN.

3.2 Blind classification with pre-fixed thresholds (Test 6)

12 runs with randomly assigned, hidden learning rates. Classification used only mean ϕ and thresholds fixed in advance (healthy $< 0.10 \leq$ intermediate $< 1.0 \leq$ stressed). Accuracy 10/12 (83%); the extremes were perfect — healthy 6/6, stressed 4/4. Both errors were 'intermediate' runs, which did not form a distinct ϕ state: training behaved as effectively binary (stable or unstable) and ϕ reflected this.

3.3 Specificity and baseline comparison (Test 7)

Six healthy runs of 300 steps each produced **zero false alarms** (ϕ never crossed 0.10 for 3 consecutive steps). A naive baseline detector (loss $> 1.5\times$ baseline median, 3 consecutive steps) fired prematurely or spuriously in 4 of 8 ramp runs (alarms at steps 60–78 for collapses occurring at steps 280–370). When collapse did occur, ϕ fired within 1–2 steps of it. Within this validation setting, ϕ 's operational profile is: silent on healthy runs, near-instantaneous on collapse. The healthy-run sample is small (6 runs); specificity should be confirmed on a larger and more varied set before being treated as established.



4. Falsified and unresolved claims

4.1 Falsified: the adaptive controller (Unified-LoRA)

The original controller (dynamic rank + LR from ϕ) does not outperform AdaLoRA on quality or stability. On a controlled toy comparison (3 seeds), AdaLoRA achieved better final loss (0.198 vs 0.226) and was the only method not diverging in the extreme regime (0/3 vs 3/3). On MRPC with 50% label noise (3 seeds), AdaLoRA F1 0.518 vs fixed LoRA 0.374 vs Unified-LoRA 0.358. The cause is structural: the controller regulates update magnitude but not direction; AdaLoRA's SVD parametrization constrains direction, which is what stabilizes hard regimes.

4.2 Unresolved: predictive lead time (Test 7)

Test 7 asked whether ϕ crosses its alarm threshold usefully *before* loss collapse (pre-registered criterion: lead ≥ 10 steps in $\geq 6/8$ runs, ≤ 1 false alarm in 6 healthy runs, and beating the naive baseline). Design: 8 runs with an exponential LR ramp (doubling period randomized per seed, so collapse timing was not known in advance) plus 6 healthy runs; all detection performed post-hoc on logged traces with causal rules (each alarm decision uses only past data).

Outcome: the predictive verdict is void due to a confound identified during analysis. The task (20 fixed pairs recycled for 400 steps) was fully memorized within ~ 20 steps; the loss then sat at exactly zero for hundreds of steps regardless of the growing LR, until an abrupt numerical blow-up. A loss pinned at zero has no dynamics: no loss-derived signal could have shown precursors on this task, so the observed lead times (-1 to -2 steps on all 4 clean runs) do not test the hypothesis. Two additional runs never collapsed within the step cap and two had corrupted baselines (median = 0), consistent with the same saturation. What the run *did* provide is the specificity observation in §3.3, which is unaffected by the confound.

A corrected protocol (Test 7 v3) replaces the recycled pairs with a non-memorizable stream (novel pairs generated at every step), keeping the loss informative throughout the run. This amendment is declared here before execution.

Test 7 — loss (blue) and ϕ (red, dotted line = threshold 0.10): loss saturates at 0, collapse is a cliff

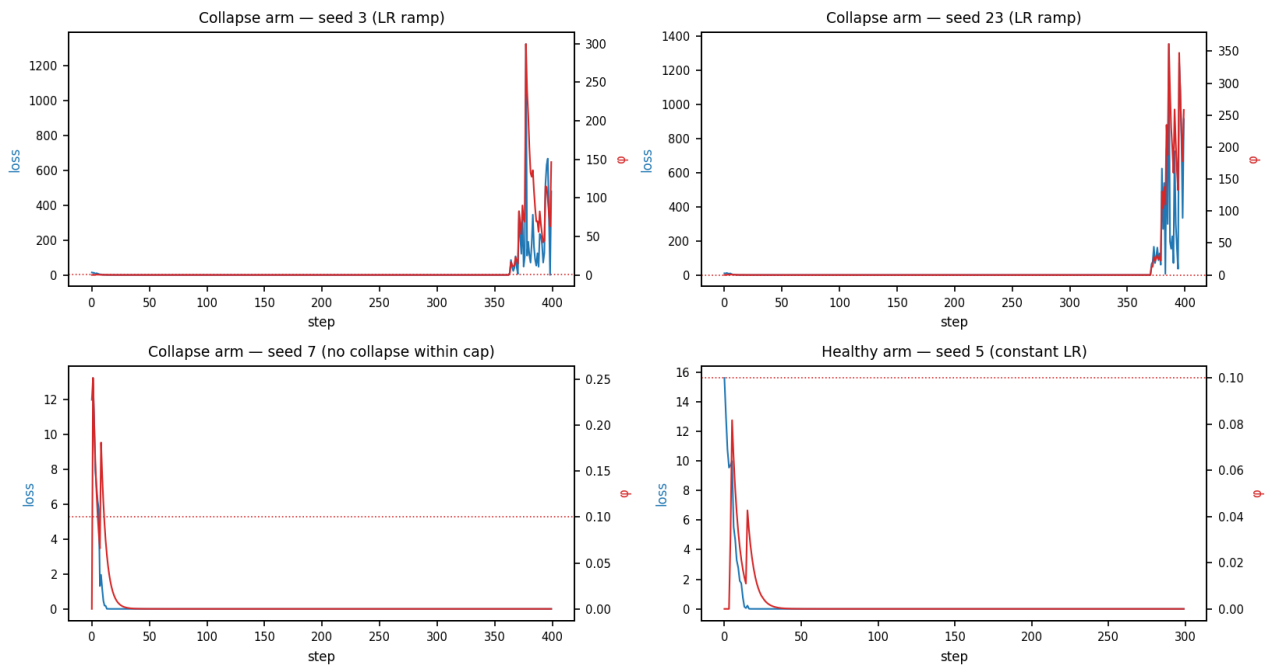


Figure: representative Test 7 traces. Loss (blue) saturates at zero after ~20 steps; collapse is an abrupt cliff; ϕ (red) fires at the cliff, 1–2 steps after the collapse criterion. Healthy arm: ϕ flat, no alarms.

5. Positioning and open questions

Related work on training stability (ZClip, SPAM, gradient-norm alerting in experiment trackers) operates on gradient norms and assumes trainer access. ϕ 's distinctive property is that it is **black-box**: computed from the loss stream alone, it can monitor fine-tuning behind managed APIs where gradients and internals are not exposed — a setting of increasing practical relevance for governance and observability of third-party training.

Open questions, in order of value: (1) predictive lead time on a non-saturating task (Test 7 v3, protocol above); (2) whether precursors absent from the loss exist in richer signals — loss variance and autocorrelation in rolling windows (computable from the same logs), gradient norms, weight drift — which requires trainer access and connects to the full controller variant of ϕ ; (3) generalization to other instability causes (small batches, noisy data, long sequences), other model families, and richer tasks; (4) operational utility: does surfacing ϕ lead a human operator to better decisions during training?

6. Reproducibility bundle

Included with this note (validation/ folder in the repository): (a) raw per-step traces — `phi_natural_stress_log.json` (Test 5), `phi_blind_diagnostic.json` (Test 6), `phi_lead_time_log.json` (Test 7, full loss/ ϕ /LR per step for all 14 runs); (b) executable scripts for Tests 1–7 with pre-registered thresholds and verdict logic in-code; (c) the reanalysis script used to audit Test 7 post-hoc without re-running. All Tinker experiments require only an API key; detection logic runs offline on the JSON logs.

Contact: Simona Vargiu — repository: github.com/Sva76/Unified-LoRa